



Force-Operated TOY ORGAN

PRADEEP G.

This toy organ for children uses a readily-available force-sensitive resistor (FSR) and an NE555 timer IC. Resistance of the FSR varies with the amount of force or pressure

being applied on its surface to play the toy organ.

Circuit and working

The circuit diagram of the force-operated toy organ is shown in Fig. 1. It is built around force sensor FSR1, timer NE555 (IC1), small loud-speaker LS1 and a few other components.

Here, FSR1 is connected between pins 6 and 7 of NE555, which is wired in astable multivibrator mode. NE555 acts as an oscillator that produces square-wave signals in audio range.

Normally, FSR1's internal resistance is above 100-kilo-ohm. When force is applied on its surface by pressing it with a finger, its resistance decreases and audio frequency or oscillation changes. So, by applying different pressures on FSR1, different audio tones are generated that can be heard in LS1.

Construction and testing

An actual-size PCB layout for the force-operated toy organ is shown in Fig. 2 and its components layout in Fig. 3. Assemble the circuit on the PCB. Connect a 9V battery across BATT.1 and FSR1 using external wires. As soon as the circuit is powered on, sound will be heard from the speaker.

Place FSR1 on a flat surface so that force can easily be applied on the active

PARTS LIST

Semiconductors:

IC1 - NE555 timer

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1 - 1-kilo-ohm

FSR1 - Force-sensitive resistor

Capacitors:

C1 - 100 μ F, 25V electrolytic

C2 - 0.04 μ F ceramic disk

C3 - 10 μ F, 25V electrolytic

C4 - 0.01 μ F ceramic disk

Miscellaneous:

BATT.1 - 9V battery

S1 - On/off switch

LS1 - 8-ohm, 0.5W loudspeaker

- 2-pin terminal connector

- Wires for circuit connections

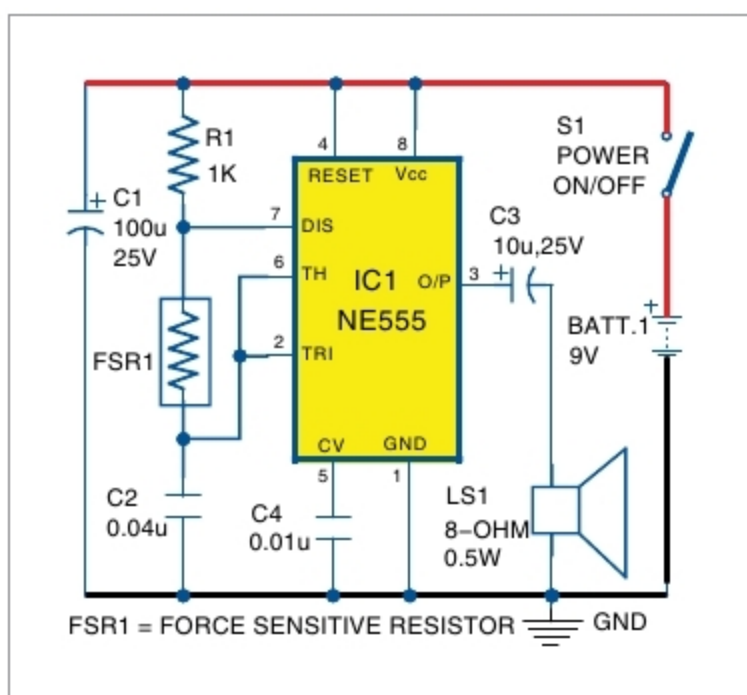


Fig. 1: Circuit diagram of force-operated toy organ

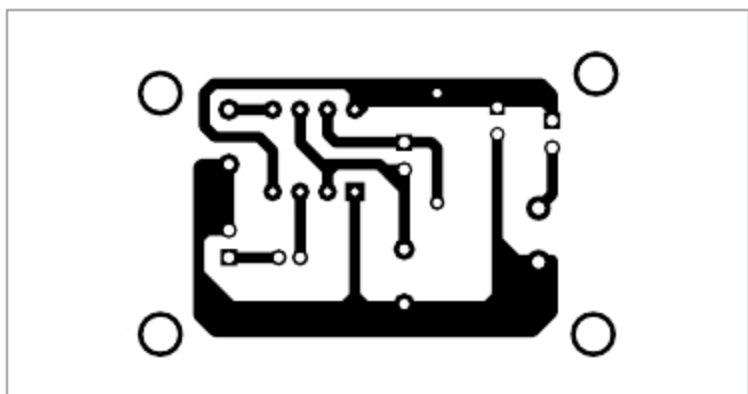


Fig. 2: Actual-size PCB layout of force-operated toy organ

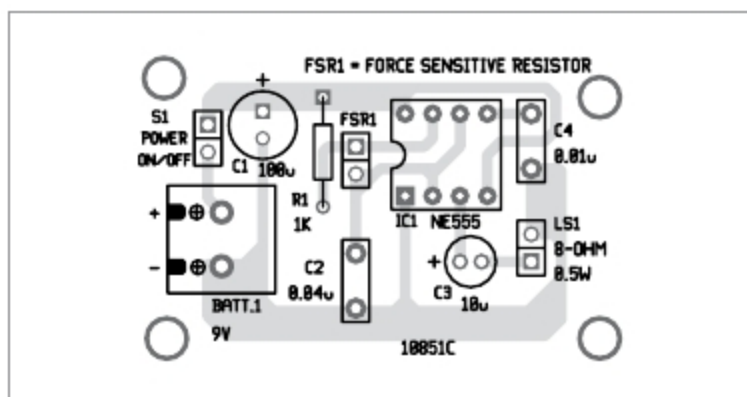


Fig. 3: Components layout for the PCB



Fig. 4: Force sensor

surface to produce various tones. The 38mm (1.5-inch) force sensor used in this project is shown in Fig. 4. It has two pins extended from the active surface for connection to a bread-board or circuit. **EFY**



Pradeep G. is an electronics hobbyist